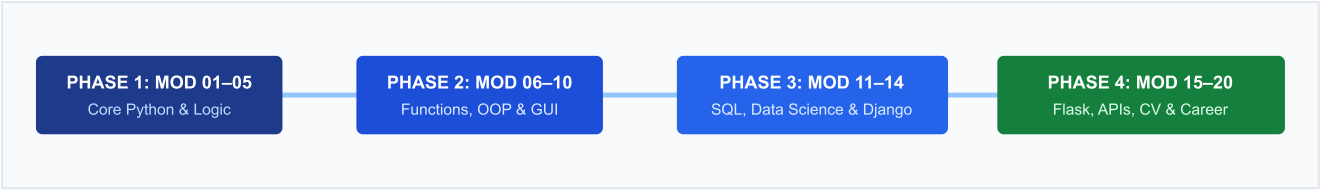
PYTHON MASTER SYLLABUS

Sequential Module-Wise Roadmap (Beginner to Full-Stack & Data Science)

20 MODULES · 270+ TOPICS
LINKEDIN SHOWCASE EDITION



MODULE 01 Python Basics, Operators & Console I/O

Topics 01 – 15

01. Literals in Python
02. Variables
03. Keywords
04. Primitive Datatypes
05. Basic String Operations
06. Type Conversion (int, float, str)
07. Type Conversion (boolean)
08. Arithmetic & Assignment Ops
09. Comparison & Logical Ops
10. Associativity & Precedence
11. Console Output: print()
12. User Input: input()
13. Getting Started with PyCharm
14. Numeric Functions
15. Foundational Coding Exercises

MODULE 02 Strings & String Formatting

Topics 01 – 05

01. Slicing in Strings
02. f-strings in Python
03. Escape Sequences
04. String Ops: Membership & strip()
05. String Ops: count, cases, start/end

MODULE 03 Lists & List Manipulation

Topics 01 – 06

01. Lists Introduction
02. Slicing, Concat, Repeat, append()
03. List Ops: extend(), remove(), pop()
04. reverse(), sort(), count(), membership
05. Numerical Operations on Elements
06. Nested Lists in Python

MODULE 04 Tuples, Sets & Dictionaries

Topics 01 – 11

01. Introducing Tuples
02. Basic Operations on Tuples
03. Lists (Mutable) vs Tuples (Immutable)
04. Introducing Sets
05. Basic Operations on Sets
06. Union, Intersection & Difference
07. Frozen Sets (frozenset)
08. Introducing Dictionaries
09. Dict Ops: Updating & Deleting
10. Working with Keys & Values
11. Shallow Copy vs Deep Copy

MODULE 05 Conditional Statements & Loops

Topics 01 – 20

01. The if Block
02. if-else Block
03. if-elif-else Statements
04. Nested if Statements
05. The Ternary Operator
06. for Loop Fundamentals
07. for Loops with Strings & Dicts
08. The range() Function
09. Total, Highest & Lowest via Loops
10. continue & break Statements
11. The while Loop
12. Infinite while Loop
13. The random Module
14. Nested Loops
15. Star Patterns Using for Loops
16. Exercise: Roll a Dice
17. Exercise: List & Loops
18. Exercise: Loops & Dictionaries
19. Number Guessing Game (Problem)
20. Number Guessing Game (Solution)

PHASE 2 (MODULES 06 – 10): Modular Programming, File I/O, RegEx, OOP & Desktop GUI **81 Sequential Lectures**

MODULE 06 Functions, Lambda & Modules

Topics 01 – 15

01. Introduction to Functions
02. User-Defined Functions
03. Returning Values from Function
04. Types of Arguments
05. Variable Positional Args (*args)
06. Variable Keyword Args (**kwargs)
07. Docstrings in Python Functions
08. Recursive Functions
09. Local & Global Variables
10. Function as an Argument
11. Lambda Anonymous Functions
12. filter() & map() with Lambda
13. Built-in Modules in Python
14. User-Defined Modules
15. What is __name__ == "__main__"

MODULE 07 File Handling, Exception Handling, JSON & Pickle

Topics 01 – 14

01. Introduction to File Handling
02. Creating Files
03. The 'w' (Write) Mode
04. Reading Files
05. Appending into a File
06. Context Manager: with Statement
07. Check if a File Exists
08. Common Issues with File Handling
09. Exceptions & try-except Block
10. Exception Handling: else & finally
11. Raising Exceptions (raise)
12. The json Module
13. The pickle Module
14. Pickle & Exception Handling

MODULE 08 Regular Expressions (RegEx Complete Track)

Topics 01 – 16

01. Intro to RegEx & Raw Strings
02. search() & match() Methods
03. Dot (.) & Character Classes
04. Special & Meta Characters Intro
05. Star (*) Meta Character
06. Plus (+) Meta Character & Example
07. Curly Braces ({}) & Example
08. Wildcard & Practical Example
09. Optional (?) Meta Character
10. Caret (^) Meta Character
11. Character Classes (Part 1 & 2)
12. Quantifiers & More Metacharacters
13. Finding Matches: findall, finditer
14. Substituting Patterns: sub()
15. The compile() Function
16. Exercise: Finding Valid Emails

MODULE 09 Object-Oriented Programming (OOP)

Topics 01 – 24

01. Introduction to OOPs
02. What are Classes & Objects
03. Creating Classes & Objects
04. What are Attributes
05. Instance Methods
06. Arguments in Instance Methods
07. Initializer & Instance Variables
08. Class Variables
09. Class Methods (@classmethod)
10. Static Methods (@staticmethod)
11. Project: Phone Book (Part 1)
12. Project: Phone Book (Part 2)
13. Inheritance Intro & Single Inheritance
14. Single Inheritance & Initialization
15. Single Inheritance: Arguments
16. Multiple Inheritance
17. Multilevel Inheritance
18. Introduction to Polymorphism
19. Operator Overloading
20. Method Overloading
21. Method Overriding
22. Encapsulation & Abstraction
23. Abstract Classes
24. Custom Exception Classes

MODULE 10 Desktop GUI Development with Tkinter

Topics 01 – 12

01. Intro to GUI & Tkinter
02. Label, title, minsize, pack()
03. Button & Change Label Text
04. The Entry Component
05. What is ttk Module
06. Quitting the Window
07. Frames & Background Colors
08. Text, Separator & Padding
09. Checkbutton Widget
10. Radiobutton Widget
11. Checkbox & Listbox
12. Spinbox Widget

Phase 3 Module	Primary Tools & Libraries	Core Deliverable / Capstone Outcome
Mod 11: PostgreSQL & Python DB	PostgreSQL, virtualenv, psycopg2	Automated Python-to-SQL CRUD Pipeline
Mod 12: Data Science & ML	Anaconda, Pandas, Seaborn, Scikit-Learn	End-to-End Regression & Statistical Evaluation
Mod 13 & 14: Django & Bootstrap	Django ORM, DTL, Admin, Bootstrap 5	Dynamic Multi-Page Blog Application with Comments

MODULE 11 PostgreSQL Databases & Python Integration

Topics 01 – 15

01. Introduction to Data
02. Introduction to Database
03. Introduction to PostgreSQL
04. Installing PostgreSQL
05. Creating a Database
06. Deleting a Database
07. Creating Table & Adding Data
08. Retrieving & Deleting DB Data
09. Setting up virtualenv
10. Installing psycopg2
11. Connecting to the Database
12. Creating Table Using Python
13. Inserting Data Using Python
14. Extracting Data from Database
15. Adding User Input to Database

MODULE 12 Data Analysis, Visualization & Machine Learning

Topics 01 – 27

01. Data Analysis Introduction
02. Installing Anaconda
03. Important Library Properties
04. Importing CSV File
05. Methods to View File
06. Data Cleaning
07. Data Formatting
08. Data Normalization (Part 1 & 2)
09. Binning (Part 1 & 2)
10. Value Counts
11. Boxplot & Scatterplot
12. Groupby Operations
13. Pivot Tables
14. Heat Map (Part 1 & 2)
15. ANOVA Statistical Test
16. Correlation (Part 1 & 2)
17. Simple Linear Regression
18. Multiple Linear Regression
19. Simple Linear Reg. in Python
20. Multiple Linear Reg. in Python
21. Regression Plot
22. Residual Plot
23. Distribution Plot (Part 1 & 2)
24. Polynomial Regression (Part 1 & 2)
25. In-Sample Evaluation
26. Mean Squared Error (MSE)
27. R-Squared (R^2 Part 1 & 2)

MODULE 13 Full-Stack Web Development with Django

Topics 01 – 30

01. Intro & Install Django
02. Django Project Structure
03. Creating & Understanding Project
04. Start Development Server
05. Django Apps, URLs & Views
06. Blog Website & Home URLs/Views
07. Dynamic Path Segments
08. Path Converters & Slugs
09. Returning HTML Response
10. Dynamic Code, Named URLs, reverse()
11. Intro, Creating & Using Templates
12. Registering & Rendering Templates
13. DTL & Variable Injection (1 & 2)
14. Filters, Tags, for, url & if Tags
15. Template Inheritance (Part 1 & 2)
16. The include Tag (1, 2 & Revisited)
17. 404 Template, Static & Global Static
18. Home Content (1 & 2) & Images
19. allposts Page (1 & 2) & View Data
20. Dynamic index.html & allposts URL
21. Fixing Individual Post Page
22. Data, Models & Migrations
23. Relationships & Adding Models
24. Admin Panel, Data, Config & Slugs
25. Querying Data from DB (1, 2 & 3)
26. Forms, Form Class & Configuration
27. CSRF Token & Saving Form Data
28. Handling Submitted Data (1 & 2)
29. Displaying & Admin Comments
30. Updating Entries with UpdateView

MODULE 14 Bootstrap Frontend Styling

Topics 01 – 08

01. Part 1: Intro to Bootstrap
02. Part 2: Adding Bootstrap to Site
03. Part 3: Bootstrap Grid System
04. Part 4: Grid Classes Part 1
05. Part 5: Grid Classes Part 2
06. Part 6: Grid Column Offset
07. Part 7: Making Images Responsive
08. Part 8: Nesting Rows & Columns

MODULE 15 Flask Web Framework, WTForms & SQLAlchemy

Topics 01 – 15

01. Intro to Flask & Project Setup
02. Routing & Response Types
03. Static Files, Dynamic URLs, Redirects
04. Jinja Templating & Base Template
05. Inheriting: Home & Dashboard (1 & 2)
06. Expense Template, Filters & Flash
07. WTForms: Register Form (Part 1 & 2)
08. Login, Expense Forms & Adding Data
09. Auth: Basics, Register & Login View
10. Auth: Logout & Protected Routes
11. Databases with SQLAlchemy
12. Model Classes, Relationships, Tables
13. Register & Login Views with DB
14. Dashboard & Add Expense with DB
15. Fetching & Deleting Expenses

MODULE 16 REST APIs Development

Topics 01 – 12

01. APIs & REST APIs Intro
02. Hello World Using REST APIs
03. Creating Model in Admin
04. Posting APIs
05. Segregating into Pages
06. Setting Authentication
07. Adding Username to Posts
08. Adding Permissions
09. Filtering Based on Usernames
10. Search Filter
11. Filtering Based on Date
12. Ordering Filters

MODULE 17 Web Scraping (BeautifulSoup) & Selenium Automation

Topics 01 – 16

01. Scraping Intro & Setup
02. Fetching Page & BeautifulSoup
03. Scraping a Real Website
04. Filtering Scraped Data
05. Functions for Fetching & Parsing
06. Functions for Filtering Data
07. Downloading Images & Wrap-Up
08. Selenium Intro & Startup Options
09. Finding Elements on Website
10. Automating Text & Dropdowns (1 & 2)
11. Automating Checkboxes & Radios
12. Uploading Files & Submit Button
13. Working with Buttons
14. Automating Navigations
15. Page Refresh & Wrap-Up

MODULE 18 Computer Vision with OpenCV

Topics 01 – 18

01. CV & OpenCV Introduction
02. Reading, Displaying, Writing Images
03. Color Spaces
04. Resizing Images
05. Cropping & Flipping Images
06. Rotating Images
07. Splitting & Merging Images
08. Basic Arithmetic on Images
09. Drawing Shapes & Text
10. Exercise: Shape & Text on Image
11. Image Thresholding
12. Blurring Techniques
13. Morphology: Erosion & Dilation
14. Morphology: Opening & Closing
15. Working with Videos
16. Video Properties & Writing Video
17. Accessing the Webcam
18. Taking Webcam Screenshots

MODULE 19 Network & Socket Programming (Chat App)

Topics 01 – 09

01. IP Address Fundamentals
02. Ports & Sockets
03. Creating a Client
04. Sending Messages
05. Using Buffer
06. Building Messaging Functionality
07. Sending Messages to Client
08. Programming Send Functionality
09. Completing Our Chat App

MODULE 20 Career Prep, Soft Skills & Added Industry Stack

Topics 01 – 15

01. Tips to Be a Good Programmer
02. Logical Thinking & Techniques (1 & 2)
03. Hirable Python Projects (Part 1 & 2)
04. Resume Mistakes & Ideal Resume
05. Portfolio Overview & Creation
06. Case Study Format (Part 1 & 2)
07. Optimizing LinkedIn Profile (1 & 2)
08. Interview Prep & STAR Framework
09. Mock Interview Flow & Rubric
10. LinkedIn Referral Outreach (1 & 2)
11. Stipend & Salary Negotiation (1 & 2)
12. Professional Roadmap (Part 1 & 2)
13. [Added] Git & GitHub Version Control
14. [Added] Decorators, Generators, Async
15. [Added] NumPy, DRF & Cloud Hosting

Ready-to-Use LinkedIn Post Caption (Copy & Paste with this PDF):

"Structured my complete Python Full-Stack & Data Science learning roadmap into 20 sequential modules (270+ topics)—covering Core Python, OOP, Tkinter, PostgreSQL, Data Analysis & ML, Django, Flask, REST APIs, Selenium, OpenCV, and Socket Programming. Sharing this syllabus checklist for anyone building their Python career path!